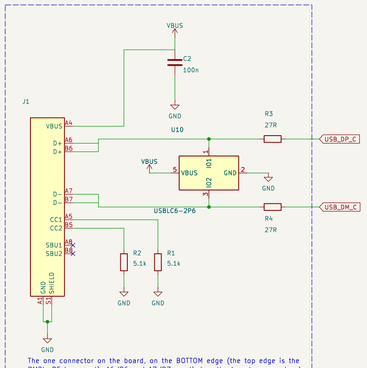
bluebuzz-rm2 — generated schematic

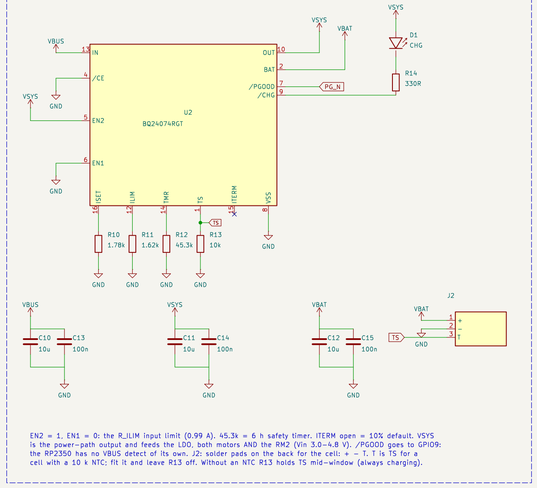
109 components / 71 nets / 6-layer. Bluetooth-audio vibrator controller: RP2350A + Raspberry Pi RM2 (CYW43439); USB-C, every other lead on back-side pads; BQ24074 charger with power path, a soft power latch (button, GPIO hold, USB), two low-side motor channels, 5 DPS368 barometers on the back (four corners and the centre, a tactile skin) on a switched 3.3 V rail, an IMU and a 1010 RGB status LED. Rails are power symbols; cross-block signals are global labels. Generated from scripts/circuit.py; rebuild with `python scripts/circuit.py && python scripts/gen_schematic.py`.

1 — USB-C and ESD



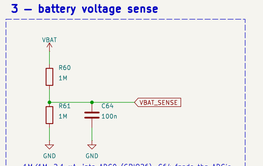
The one connector on the board, on the BOTTOM edge (the top edge is the RM2's RF keep-out). A6/B6 and A7/B7 are tied or the board enumerates in one cable orientation only; 5.1k on each CC. The USBLC6-2P6 clamps VBUS and protects D+/D−; 27 R in each line at the RP2350 is the reference design's.

2 — BQ24074 charger with power path



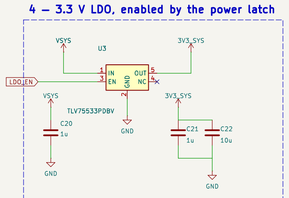
EN2 = 1, EN1 = 0: the R_ILIM input limit (0.99 A). 45.3k = 6 h safety timer. ITERM open = 10% default. VSYS is the power-path output and feeds the LDO, both motors AND the RM2 (Vin 3.0–4.8 V). /PGOOD goes to GPIO9; the RP2350 has no VBUS detect of its own. J2: solder pads on the back for the cell: + — T. T is TS for a cell with a 10 k NTC; fit it and leave R13 off. Without an NTC R13 holds TS mid-window (always charging).

3 — battery voltage sense



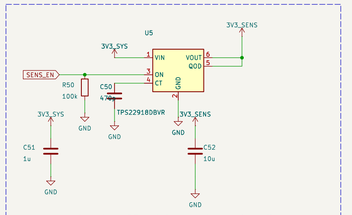
1M/1M, 2.1 uA, into ADC0 (GPIO26). C64 feeds the ADC's sampling charge; the RP2350's ADC input is 500 ohm-ish during a sample, so oversample.

4 — 3.3 V LDO, enabled by the power latch



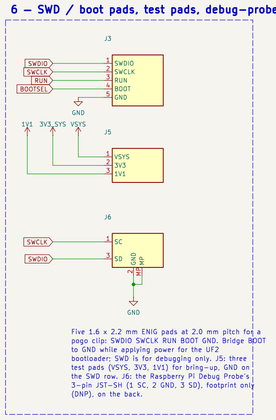
500 mA, 25 uA Iq. Feeds the RP2350, the flash, the sensors, the IMU and the RM2's VDDIO; off (EN low) is the system's off state. The motors and the RM2's Vin take VSYS directly.

5 — switched sensor rail



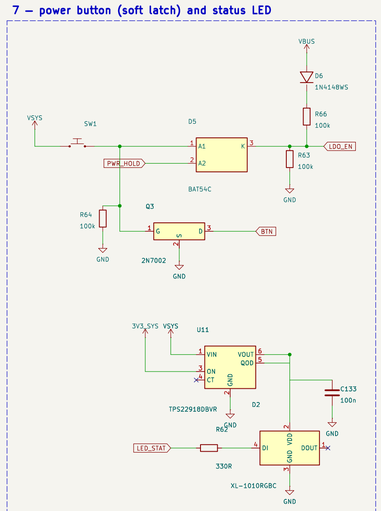
The switched 3.3 V rail for the tactile array (block 6): the sensors draw little, but gating them lets firmware power-cycle the array and keeps the rail off in sleep.

6 — SWD / boot pads, test pads, debug-probe connector (back)



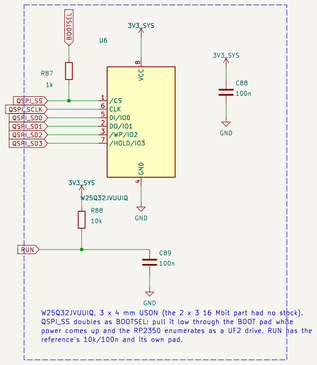
Five 1.6 x 2.2 mm ENIG pads at 2.0 mm pitch for a pogo clip: SWDIO SWCLK RUN BOOT GND. Bridge BOOT to GND while applying power for the UF2 bootloader; SWD is for debugging only. J5: three test pads (VSYS, 3V3, 1V1) for bring-up. GND on the SWD row. J6: the Raspberry Pi Debug Probe's 3-pin JST-SH (1 SC, 2 GND, 3 SD), footprint only (DNP), on the back.

7 — power button (soft latch) and status LED



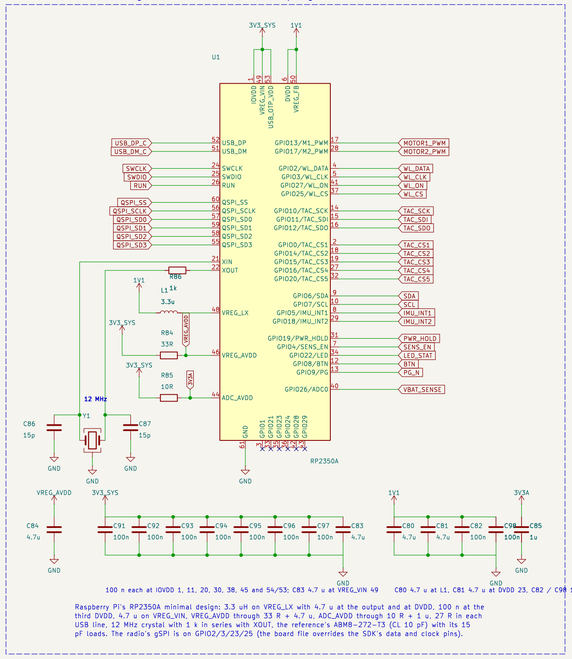
Press SW1: VSYS reaches the LDO's EN through D5 and 3V3 comes up; the RP2350 boots and drives PWR_HOLD (GPIO19) high, so the board stays on after release. Firmware reads the button on GPIO8 (Q3, active low, internal pull-up) and switches off by driving PWR_HOLD low. R63 holds EN off; R64 discharges the gate. USB in (VBUS through D6 and R66) also holds EN on, so flashing and SWD need no button. Charging runs while off. Status: a 1010 addressable RGB LED (WS2812B type) on GPIO22 through 330 R; its VSYS supply goes through U11 (a second TPS22918, ON = 3V3_SYS), so it draws nothing while the system is off.

8 — QSPI flash, reset and boot



W25Q32JVUUIQ, 3 x 4 mm USON (the 2 x 3 16 Mbit part had no stock). QSPI_SS doubles as BOOTSEL: pull it low through the BOOT pad while power comes up and the RP2350 enumerates as a UF2 drive. RUN has the reference's 10k/100n and its own pad.

9 — RP2350A: core regulator, clock, USB, decoupling

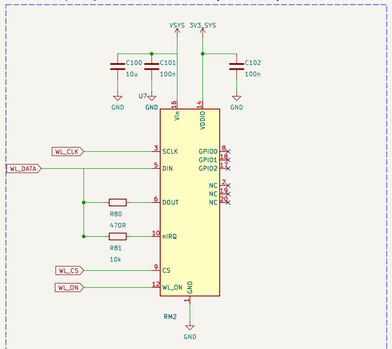


Raspberry Pi's RP2350A minimal design: 3.3 uH on VREG_LX with 4.7 u at the output and at DVDD. 100 n at the third DVDD. 4.7 u on VREG_VIN, VREG_AVDD through 33 R + 4.7 u. ADC_AVDD through 10 R + 1 u. 27 R in each USB line. 12 MHz crystal with 1 k in series with XOUT, the reference's ABM8-272-T3 (CL 10 pF) with its 15 pF loads. The radio's gSPI is on GPIO2/3/23/25 (the board file overrides the SDK's data and clock pins).

100 n each at IOVDD 1, 11, 20, 30, 38, 45 and 54/53; C83 4.7 u at VREG_VIN 49

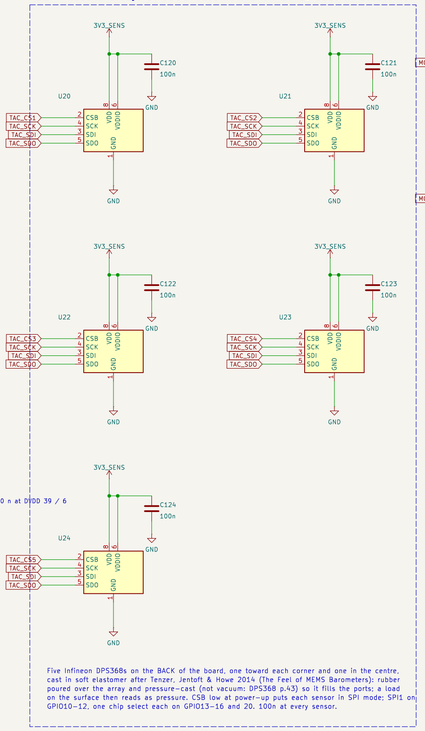
C80 4.7 u at L1, C81 4.7 u at DVDD 23, C82 / C98 100 n at DVDD 39 / 6

10 — Raspberry Pi Radio Module 2 (CYW43439)



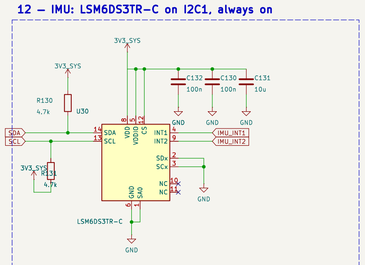
RMC20452T: Wi-Fi 4 + Bluetooth 5.2 Classic/LE, PCB antenna, modular certification. Vin 3.0–4.8 V from VSYS (271 mA TX peaks stay off the LDO), VDDIO 3V3. gSPI: GPIO3 clock, GPIO2 data (DIN direct, DOUT via 470 R. nIRQ via 10 k), GPIO25 CS, GPIO27 WL_ON on pads 12 and 13 (WL_REG_ON and BT_REG_ON). The datasheet's RF keep-out — 5 mm into the body, 10.5 mm either side, 13.5 mm beyond — is a rule area on every layer; the module's antenna end is the board edge.

11 — tactile array: DPS368 barometers on SPI1, cast in elastomer



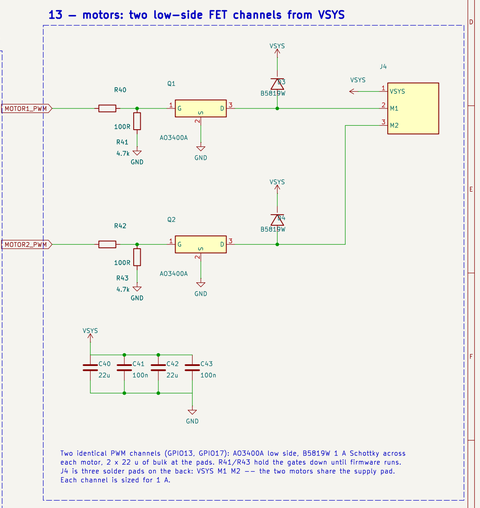
Five Infineon DPS368s on the BACK of the board, one toward each corner and one in the centre, cast in soft elastomer after Tenzer, Jentoft & Howe 2014 (The Feel of MEMS Barometers): rubber poured over the array and pressure-cast (not vacuum; DPS368 p.43) so it fills the ports; a load on the surface then reads as pressure. CSB low at power-up puts each sensor in SPI mode; SPI1 on GPIO10–12, one chip select each on GPIO13–16 and 20. 100n at every sensor.

12 — IMU: LSM6DS3TR-C on I2C1, always on



ST LSM6DS3TR-C six-axis IMU on I2C1 (GPIO6/7, 4.7k pull-ups to the always-on rail); CS high selects I2C, SA0 low gives address 0x6A; the aux I2C (SDx/SCx) is unused and tied to GND. INT1/INT2 to GPIO5/18 for wake-on-motion, tap and tilt. 100n + 10u on VDD, 100n on VDDIO.

13 — motors: two low-side FET channels from VSYS



Two identical PWM channels (GPIO13, GPIO17): AO3400A low side, B5819W 1 A Schottky across each motor. 2 x 22 u of bulk at the pads. R41/R43 hold the gates down until firmware runs. J4 is three solder pads on the back: VSYS M1 M2 — the two motors share the supply pad. Each channel is sized for 1 A.

Power flags — the ERC root for each rail



No copper, no netlist entry: every rail is fed through a pin the symbols declare passive, so ERC needs a driver per rail.

generated from scripts/circuit.py by gen_schematic.py
RP2350A + RM2 vibrator controller, DPS368 tactile array, LSM6DS3TR-C IMU
sequoia-hope
Sheet: /
File: bluebuzz-rm2.kicad_sch
Title: bluebuzz-rm2
Size: User | Date: 2026-10-18 | **Rev: A**
KiCad E.D.A. 9.0.8 | Id: 1/1